

Definitions and standards

You will find in this document the glossary for the items displayed in the graphs. Please make sure you refer to the definition of each aggregate item in the relevant context of use (e.g., harvested crops, diet composition). Some items are labelled “FAO definition”; in those cases, the data come from FAOstat (www.fao.org) and the definitions follow FAOstat’s own classification. This document provides one glossary table per application tab.

AGGREGATED PRODUCTS USED TO RUN THE MODEL	
Item	Definition
Aquatic animal	Freshwater, demersal, pelagic and other marine fish; crustaceans, cephalopods and other molluscs; meat aquatic mammals and other aquatic animals
Bovine	Bovine meat
Small ruminant	Sheep and goats meat
Pork	Pork meat
Poultry	Poultry meat
Eggs	Eggs
Dairy	Dairy products
Grass	From permanent meadows and pastures
Grass-like forage	From temporary meadows and pastures (mixed grass and ray-grass)
Other forages	Cultivated forages (alfalfa, beets, legumes, maize, etc.).
Fibers	Jute, jute-like fibres, soft-fibres other, sisal, abaca, hard fibres other, tobacco, rubber and seed cotton
Roots and Tuber	Potatoes, cassava, sweet potatoes, yams and other roots
Fruits & vegetables	Tomatoes, onions, vegetables other, oranges, mandarines, lemons, limes, grapefruit, citrus other, bananas, plantains, apples, pineapples, dates, grapes and other fruits
Maize	Maize
Wheat	Wheat
Rice	Rice, paddy equivalent
Other cereals	Barley, rye, oats, millet, sorghum and other cereals
Pulses	Beans, peas and other pulses
Soyabeans	Soyabeans
Soyabean Cake	Soyabean cake
Soyabean Oil	Soyabean oil
Sunflowerseed	Sunflowerseed
Sunflowerseed Cake	Sunflowerseed cake
Sunflowerseed Oil	Sunflowerseed oil
Rape and Mustardseed	Rape and mustardseed
Rape and Mustard Cake	Rape and mustard cake
Rape and Mustard Oil	Rape and mustard oil
Other Oilcrops	Groundnuts (shelled eq), coconuts – incl copra, sesameseed, olives and other oilcrops
Cake Other Oilcrops	Other oilcrops cake
Oil Other Oilcrops	Other oilcrops oil
Oilpalm fruit	Oilpalm fruit
Palmkernel Cake	Palm kernel cake
Palm Products Oil	Palm oil and palmkernel oil
Sugar	Sugar cane, sugar beet (sugar in equivalent sugar cane and beet)
Other plant products	Nuts, coffee, cocoa beans, tea, pepper, pimento, cloves, spices, other
Crop residues	Stover
Other products	Meat other, offals edible, fats animals raw, honey, meat meal, aquatic plants
Occasional	Food leftovers, cut-and-carry, forages and legumes, roadside grasses

Tab n°1 : “ASSUMPTIONS”

Item	Definition
Food supply (kcal/capita/day) (FAO's definition)	Refers to the total amount of food available for human consumption expressed in kilocalories (kcal) per capita per day. Caloric content is derived by applying the appropriate food composition factors to the quantities of the commodities.
Cereals	Wheat, maize, rice, mil, sorghum and all the other cereals
Dairy	Milk and milky products
Oil	Soybean oil, rapeseed oil, sunflower oil, palm and palm kernel oil and other oilseed (copra, peanut, sesame, flax, etc.)
Pulse	Peas, beans, lentils and other
Vegetables and fruits	Tomatoes, onions, oranges, mandarins, lemons, grapefruit, bananas, plantains, apples, pineapples, dates, grapes and other fruits and vegetables
Roots and Tuber	Potatoes, cassava, sweet potatoes, yams and other roots
Meat, eggs and fish	Monogastric and ruminant meat, eggs and fish
Sugar	Sugar and sugar plants (sugarcane, sugar beets, sugar in cane or beet equivalent)
Other	All the other products include stimulants (tobacco, rubber...)

Tab n°2 : “BIG PICTURE”

Item	Definition
Land uses	
Cropland (FAO's definition)	Land used for cultivation of crops. The total of areas under "Arable land" and "Permanent crops". Arable land : Land used for cultivation of crops in rotation with fallow, meadows and pastures within cycles of up to five years. The total of areas under "Temporary crops," "Temporary meadows and pastures," and "Temporary fallow." Arable land does not include land that is potentially cultivable but is not cultivated. Permanent crops : Land cultivated with long-term crops which do not have to be replanted for several years (such as cocoa and coffee), land under trees and shrubs producing flowers (such as roses and jasmine), and nurseries (except those for forest trees, which should be classified under "Forestry"). Permanent meadows and pastures are excluded from Permanent crops.
Land under perm. meadows and pastures (FAO's definition)	Land used permanently (five years or more) to grow herbaceous forage crops through cultivation or naturally (wild prairie or grazing land). Permanent meadows and pastures on which trees and shrubs are grown should be recorded under this heading only if the growing of forage crops is the most important use of the area. Measures may be taken to keep or increase productivity of the land (i.e., use of fertilizers, mowing or systematic grazing by domestic animals.) This class includes: • Grazing in wooded areas (agroforestry areas, for example); • Grazing in shrubby zones (heath, maquis, garigue); • Grassland in the plain or low mountain areas used for grazing: land crossed during transhumance where the animals spend a part of the year (approximately 100 days) without returning to the holding in the evening: mountain and subalpine meadows and similar; • Steppes and dry meadows used for pasture.
Forest land (FAO's definition)	Land spanning more than 0.5 hectares with trees higher than 5 meters and a canopy cover of more than 10 per cent, or trees able to reach these thresholds in situ. Excludes land that is predominantly under agricultural or urban land use (more on FAOstat website).
Agricultural land	“Cropland” and “Land under perm. meadows and pastures”

Balance

Production (FAO's definition)	Figures relate to the total domestic production whether inside or outside the agricultural sector, i.e. it includes non-commercial production and production from kitchen gardens. Unless otherwise indicated, production is reported at the farm level for crop and livestock products (i.e. in the case of crops, excluding harvesting losses) and in terms of live weight for fish items (i.e. the actual ex-water weight at the time of the catch). All data shown relate to total meat production from both commercial and farm slaughter. Data are expressed in terms of dressed carcass weight, excluding offal and slaughter fats. Production of beef and buffalo meat includes veal; mutton and goat meat include meat from lambs and kids; pig meat includes bacon and ham in fresh equivalent. Poultry meat includes meat from all domestic birds and refers, wherever possible, to ready-to-cook weight.
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Import quantity (FAO's definition)	
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Domestic supply (FAO's definition)	Supply for domestic utilization is defined as : Production + Imports - Exports
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Export quantity (FAO's definition)	
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Food (FAO's definition)	Data refer to the total amount of the commodity available as human food during the reference period. Data include the commodity in question, as well as any commodity derived therefrom as a result of further processing. Food from maize, for example, comprises the amount of maize, maize meal and any other products derived therefrom available for human consumption. Food from milk relates to the amounts of milk as such, as well as the fresh milk equivalent of dairy products.
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Feed (FAO's definition)	Data refer to the quantity of the commodity in question available for feeding to the livestock and poultry during the reference period, whether domestically produced or imported.
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Seed (FAO's definition)	Data include the amounts of the commodity in question set aside for sowing or planting (or generally for reproduction purposes, e.g. sugar cane planted, potatoes for seed, eggs for hatching and fish for bait, whether domestically produced or imported) during the reference period. Account is taken of double or successive sowing or planting whenever it occurs. The data of seed include also, when it is the case, the quantities necessary for sowing or planting the area relating to crops harvested green for fodder or for food. (e.g. green peas, green beans, maize for forage) Data for seed element are stored in tonnes (t). Whenever official data were not available, seed figures have been estimated either as a percentage of supply (e.g. eggs for hatching) or by multiplying a seed rate with the area under the crop of the subsequent year.
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Losses (FAO's definition)	Amount of the commodity in question lost through wastage (waste) during the year at all stages between the level at which production is recorded and the household, i.e. storage and transportation. Losses occurring before and during harvest are excluded. Waste from both edible and inedible parts of the commodity occurring in the household is also excluded. Quantities lost during the transformation of primary commodities into processed products are taken into account in the assessment of respective extraction/conversion rates. Distribution wastes tend to be considerable in countries with hot humid climate, difficult transportation and inadequate storage or processing facilities. This applies to the more perishable foodstuffs, and especially to those which have to be transported or stored for a long time in a tropical climate. Waste is often estimated as a fixed percentage of availability, the latter being defined as production plus imports plus stock withdrawals
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Other uses (non-food) (FAO's definition)	Data refer to quantities of commodities used for non-food purposes, e.g. oil for soap. In order not to distort the picture of the national food pattern quantities of the commodity in question consumed mainly by tourists are included here (see also "Per capita supply"). In addition, this variable cover pet food.
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Tab n°3 : “CROPS”

Item	Definition
Harvested crops aggregates	
Millet and Sorghum	Millet and Sorghum
Oilcrops	Rapeseed, sunflower, peanut, sesame and other annual oilcrops (unless soybeans)
Other	Fruits and vegetables and fibers
Other Cereals	Barley, rye, oats and other
Pulse and Soybeans	Beans, chickpea, lentils, soybeans and other pulses
Rice	Rice
Roots and Tuber	Potatoes, cassava, sweet potatoes, yams and other roots
Perennial plants and stimulants	Tea, cocoa, coffee, oilpalm, sugar cane, tobacco, rubber and other
Wheat	Wheat
Maize	Maize

Land uses items (in ha)

Harvested area (FAO's definition)	Data refer to the area from which a crop is gathered. Area harvested, therefore, excludes the area from which, although sown or planted, there was no harvest due to damage, failure, etc. It is usually net for temporary crops and sometimes gross for permanent crops. Net area differs from gross area insofar as the latter includes uncultivated patches, footpaths, ditches, headlands, shoulders, shelterbelts, etc. If the crop under consideration is harvested more than once during the year as a consequence of successive cropping (i.e. the same crop is sown or planted more than once in the same field during the year), the area is counted as many times as harvested. On the contrary, area harvested will be recorded only once in the case of successive gathering of the crop during the year from the same standing crops. With regard to mixed and associated crops, the area sown relating to each crop should be reported separately. When the mixture refers to particular crops, generally grains, it is recommended to treat the mixture as if it were a single crop; therefore, area sown is recorded only for the crop reported.
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Balance items : definitions in tab n°2

Tab n°4 : “LIVESTOCK”

Item	Definition
Livestock for area items (in ha)	
Beef cattle	Area allocated to beef cattle (meat production)
Dairy	Area allocated to dairy cattle and dairy small ruminants
Small meat ruminant	Area allocated to small meat ruminants (meat production)
Animals stock items (in head or TLU)	
TLU	Tropical Livestock Unit
Beef cattle (TLU)	All males and females that are not dairy will not become dairy
Dairy cattle (TLU)	Productive dairy cows, goats and sheeps and replacement heifers
Sheep and goats meat (TLU)	All males and females that are not dairy will not become dairy

Pigs (head)	Pig and sows
Poultry meat (head)	All poultry intended to produce meat
Poultry eggs (head)	All poultry intended to produce eggs
Animals stock per system items (TLU)	
Beef cattle	All males and females that are not dairy will not become dairy
Dairy cattle	Productive dairy cows and replacement heifers
Small dairy ruminants	Productive dairy goats and sheeps and replacement heifers
Small meat ruminants	All males and females that are not dairy will not become dairy

Livestock system	
MixedArid	Mixed crop–livestock farming in arid areas, combining grazing with crop-derived feed resources under strong climatic constraints
MixedNonArid	Mixed crop–livestock farming in non-arid areas, with tighter crop–animal integration supported by more reliable feed availability
PastoralArid	Extensive pastoralism in arid areas, relying on natural rangelands and high herd mobility to track feed and water
PastoralNonArid	Pastoral livestock production in non-arid areas, based on rangelands with seasonal mobility and generally lower climatic stress
Specific	Others livestock systems (mostly urban) Other, often urban or peri-urban livestock systems, typically more specialized and dependent on purchased feeds and market access

Ruminants products (in tons or kcal)	
Bovine meat from beef cattle	Beef from meat livestock
Bovine meat from dairy cattle	Beef from reform dairy cows
Small ruminants meat from meat animals	Small ruminants meat from livestock for meat
Small ruminants meat from dairy cattle	Small ruminants meat from reform dairy small ruminants
Dairy	Dairy from cattle and small ruminants

Balance items : definitions in tab n°2

Tab n°5 : “LAND USE”	
Item	Definition
Land use items : definitions in tab n°2	

Tab n°6 : “BALANCE”	
Item	Definition
Balance items : definitions in tab n°2	

Tab n°7 : “DEPENDANCY”	
Item	Definition
Balance items : definitions in tab n°2	

TAB N°8 : “EMISSIONS”

Item	definition
Total emissions in CO₂e	
Energy for crops	Emissions linked to energy use in crop production (e.g., fuel and electricity used for field operations/irrigation and other crop-related energy uses)
Enteric	Emissions from enteric fermentation in ruminant livestock (mainly methane)
Production and application of fertilizer and pesticides	Emissions associated with the upstream production/manufacturing of mineral fertilisers and pesticides and their application in the field
Livestock other	Manure-related sources such as storage/management and manure deposited/applied
Methane (CH₄) emissions in CO₂e	
Manure management	Methane emitted during manure storage and treatment
Enteric	Methane from enteric fermentation
Energy	Methane emissions linked to energy use/supply in the agricultural system
On-farm energy use	The on-farm component of energy-related methane emissions
Nitrous oxide (N₂O) emissions in CO₂e	
Residues Nitrogen Fertilizer applied	Emissions arising from nitrogen in crop residues returned to soils (decomposition/mineralization leading to direct/indirect N ₂ O)
Synthetic Nitrogen Fertilizer applied	Emissions resulting from application of synthetic nitrogen fertilizer to soils
Manure Nitrogen Fertilizer applied	Emissions resulting from application of animal manure to soils as an organic nitrogen fertilizer
Left on pasture	Emissions from manure and urine deposited directly on pasture/rangeland by grazing animals
Energy	Emissions linked to energy use/supply in the agricultural system
On-farm energy use	The on-farm component of energy-related N ₂ O emissions
Manure management	Emissions from manure management systems (storage/treatment)

DATA SOURCES

Yield projections:

Mueller, N. D., Gerber, J. S., Johnston, M., Ray, D. K., Ramankutty, N., & Foley, J. A. (2012). Closing yield gaps through nutrient and water management. *Nature*, 490(7419), 254–257. <https://doi.org/10.1038/nature11420>

Müller, C., & Robertson, R. D. (2014). Projecting future crop productivity for global economic modeling. *Agricultural Economics*, 45, 37–50. <https://doi.org/10.1111/agec.12088>

Monfreda, C., N. Ramankutty, and J. A. Foley, Farming the Planet. Part 2: The Geographic Distribution of Crop Areas and Yields in the Year 2000, Glob. Biogeochem. Cycles, doi:10.1029/2007GB002947, 2008.

Demographic projection:

United Nations Population Division. (2025). Population Division Data Portal. <https://population.un.org/dataportal/data/indicators/49/locations>

Livestock efficiency:

Herrero, M., Havlík, P., Valin, H., Notenbaert, A., Rufino, M. C., Thornton, P. K., Blümmel, M., Weiss, F., Grace, D., & Obersteiner, M. (2013). Biomass use, production, feed efficiencies, and greenhouse gas emissions from global livestock systems. *Proceedings of the National Academy of Sciences*, 110(52), 20888–20893. <https://doi.org/10.1073/pnas.1308149110>

Diet projections:

Alexandratos, N., & Bruinsma, J. (2012). *World agriculture towards 2030/2050: The 2012 revision* [Report]. <https://doi.org/10.22004/AG.ECON.288998>

Le Mouél, C., de Lattre-Gasquet, M., & Mora, O. (Eds.). (2018). *Land use and food security in 2050: A narrow road* (Agrimonde-Terra). Éditions Quae. <https://doi.org/10.35690/978-2-7592-2880-5>